



Time and Space Complexity Variants

Time Complexity



Example 1:

```
def sum(a: List[int], b: List[int])  
-> int:  
    n,m = len(a), len(b)  
    total = 0  
  
    for i in range(n):  
        total += a[i]  
  
    for i in range(m):  
        total += b[i]  
  
    return total
```

Time Complexity: _____
Space Complexity: _____

Example 2:

```
def get_sum_and_product(a:
List[int]) -> List[int]:
    n = len(a)
    total, product = 0, 1

    for i in range(n):
        total += a[i]

    for i in range(n):
        product *= a[i]

    return [total, product]
```

Time Complexity : _____

Space Complexity: _____

Example: 3

```
def sum(mat: List[List[int]]) ->
int:
    '''
    Example:
    mat = [[3,2,7],[2,6,8],[5,1,9]]
    '''

    total = 0
    n,m = len(mat),len(mat[0])

    for i in range(n):
        for j in range(m):
            total += mat[i][j]

    return total
```

Time Complexity : _____

Space Complexity: _____

Example 4:

```
def sort(arr: List[int]) -> None:
    for i in range(1, len(arr)):
        key, j = arr[i], i - 1

        while j >= 0 and key <
arr[j]:
            arr[j + 1] = arr[j]
            j -= 1

        arr[j + 1] = key
```

Time Complexity : _____

Space Complexity: _____

Example 5:

```
def search(arr: List[int], target: int) ->
int:
    left, right = 0, len(arr) - 1

    while left <= right:
        mid = left + (right - left) // 2

        if arr[mid] == target:
            return mid

        if arr[mid] < target:
            left = mid + 1

        else:
            right = mid - 1

    return -1
```

Time Complexity : _____

Space Complexity: _____

Example 6:

```
def function(n: int, k: int) ->  
None:  
    '''  
    Assume k >= 2  
    '''  
    count, i = 0, 1  
  
    while i <= n:  
        count += 1  
        i *= k
```

Time Complexity : _____
Space Complexity: _____

Example 7:

```
def function(n):  
    count, i = 0, n  
  
    while i > 0:  
        for j in range(i):  
            count += 1  
  
        i //= 2
```

Time Complexity : _____

Space Complexity: _____

Example 8:

```
def function(n: int) -> None:
    p, i = 0, 1

    while p <= n:
        p += i
        i += 1
```

Time Complexity : _____

Space Complexity: _____

Example 9:

```
def function(n : int) -> None:
    count, i = 0, 2

    for j in range(n):
        if i >= n:
            break

    count += 1
    i *= i
```

Time Complexity : _____

Space Complexity: _____

Example 10:

```
def add(arr: List[int]) -> int:
    c = int(10e6) # c = 1,000, 000

    total = 0

    for i in range(min(c, len(arr))):
        total += arr[i]

    return total
```

Time Complexity : _____

Space Complexity: _____

Example 11:

```
def function(n: int) -> None:
    count = i = 0

    while i <= n:
        i *= 2
        count += 1
```

Time Complexity : _____

Space Complexity: _____

Space Complexity



Example 1:

```
def add(arr : List(int)) -> int:  
    total = 0  
  
    for num in arr:  
        total += num  
  
    return total
```

Example 2:

```
def add(a: int, b: int, c: int) -> int:  
    d = a + b + c  
  
    return d
```

Example 3:

```
def first_repeated_number(arr: List[int]) -> int:
    visited = set()

    for num in arr:
        if num in visited:
            return num
        visited.add(num)

    return -1
```

Example 4:

```
def first_repeated_character(arr: List[str]) -> str:
    visited = set()

    for char in arr:
        if char in visited:
            return char

        visited.add(char)

    return '-1'
```

Example 5:

```
def min_distance(word1: str, word2: str) -> int:
    n = len(word1)
    m = len(word2)

    memo = [[-1] * (m + 1) for _ in range(n + 1)]

    for i in range(m + 1):
        memo[n][i] = m - i

    for i in range(n + 1):
        memo[i][m] = n - i

    for i in range(n - 1, -1, -1):
        for j in range(m - 1, -1, -1):
            if word1[i] == word2[j]:
                memo[i][j] = memo[i + 1][j + 1]
            else:
                memo[i][j] = min(memo[i + 1][j + 1], \
                                   memo[i + 1][j], memo[i][j + 1] + 1)

    return memo[0][0]
```

Quote of The Day

If you can't explain it simply, you don't understand it well enough.

- Albert Einstein